

Harish Prabhu

Robotics Software Engineer – Entry Level

Vilnius, Lithuania | +370 68164630 | harishprabhu3007@gmail.com |

<https://www.linkedin.com/in/harishprabhu3007/> | <https://github.com/Harishprabhu30/>

SUMMARY

MSc AI Systems student graduating June 2026 with hands-on experience in robotics perception, SLAM systems, and sensor-based platforms. Built and evaluated multi-sensor localization pipelines using ROS2 and NVIDIA Isaac Sim, working with camera, LiDAR, IMU, and encoder data. Experienced in system testing, debugging, and performance evaluation. Seeking entry-level roles in robotics software, SLAM, and autonomous systems.

SKILLS

- **Programming & OS:** Python, C++, SQL, Linux (Ubuntu)
- **Robotics:** ROS2, SLAM (EKF, Visual SLAM), sensor fusion, localization & mapping, Isaac Sim, Gazebo, basic robot kinematics, coordinate transformations, control theory fundamentals
- **Computer Vision:** OpenCV, feature extraction, visual perception concepts
- **Embedded & Hardware:** ESP32, IMU, radar, sensors, UART/I2C/SPI, RTOS basics, soldering & debugging
- **Data & Tools:** NumPy, Pandas, Matplotlib, rosbags, Git, Excel, CAD - Solidworks

TECHNICAL EXPERIENCE

SLAM for Autonomous Robots (Master's Thesis)

Sept. 2024 - May 2026, Ongoing

Vilnius Tech University, Lithuania

- Designed a modular SLAM evaluation framework using ROS2 and NVIDIA Isaac Sim with configurable environments (lighting, trajectories, sensor noise).
- Integrated camera, LiDAR, IMU, and wheel encoders on a robotic platform (Nova Carter), applying coordinate transformations and basic kinematics.
- Developed ROS2 pipelines for dataset recording and controlled estimator execution (wheel odometry, EKF, ORB-SLAM3).
- Implemented Python-based evaluation framework to compute ATE, RPE, yaw error, drift rate, and analyze system performance.
- Investigated synchronization issues, estimator drift, and real-time system behavior in robotics pipelines.
- Preparing results for IEEE conference submission (May 2026).

GitHub: <https://github.com/Harishprabhu30/ekf-slam-thesis>

AI Fellowship – Radar-Based SLAM Internship

April - May 2026

SustainLivWork Program; Tampere University, Finland

- Selected for an international AI fellowship programme focused on applied research and industry-relevant projects.
- Working on radar-based SLAM for autonomous heavy machines in a real laboratory environment.
- Performing data collection, validation, and system-level evaluation for robust localization.
- Collaborating with engineers and researchers on experimental setup, testing workflows, and documentation.

Human Motion Analysis Using IMU Sensors

- Designed structured IMU data acquisition using RTOS-based task scheduling concepts.
- Processed and analyzed time-series sensor data for motion pattern evaluation.
- Performed calibration and noise analysis to improve measurement reliability.

Embedded Systems & Electronics Projects

- Built automatic street-light system using LDR and microcontroller.
- Implemented control logic and performed hardware testing, wiring, and debugging.

EDUCATION

Master of Science in Artificial Intelligence Systems

Sept 2024 – June 2026

Vilnius Tech University, Lithuania

- Focus areas: Robotics, Autonomous Systems, SLAM, Sensor Data Processing

Bachelor of Engineering in Computer Science Engineering

June 2018 – Sept 2022

Dr. T. Thimmaiah Institute of Technology, India

- Thesis: Glaucoma Detection using Machine Learning

CERTIFICATION

- CCNA – Cisco Certified Network Associate (TCP/IP & Networking Essentials)
- Real-Time AI & ML Applications – National Conference Participation

LANGUAGES

- English – Professional working proficiency
- German – A2 (currently improving)
- Lithuanian – A1 (currently learning)
- Tamil – Native
- Hindi - Fluent